

ELECTRON TRANSPORT CHAIN

INTRODUCTION TO ETC:

- Electron Transport Chain (ETC) is also called Respiratory Chain.
- The final steps in the overall oxidation of food stuffs (carbohydrate, fat and amino acids) result in formation of NADH and FADH₂.
- The electron transport chain (ETC) oxidizes NADH and FADH₂ by transferring electrons (reducing equivalents) by a series of oxidation reduction reactions to O₂, the terminal electron acceptor. In the presence of O₂, the ETC converts reducing equivalents into energy, (ATP) by oxidative phosphorylation.

LOCALIZATION OF THE ELECTRON TRANSPORT CHAIN:

The electron transport chain is present in the inner mitochondrial membrane (Figure 10.1). The enzymes of the electron transport chain are embedded in the inner membrane.

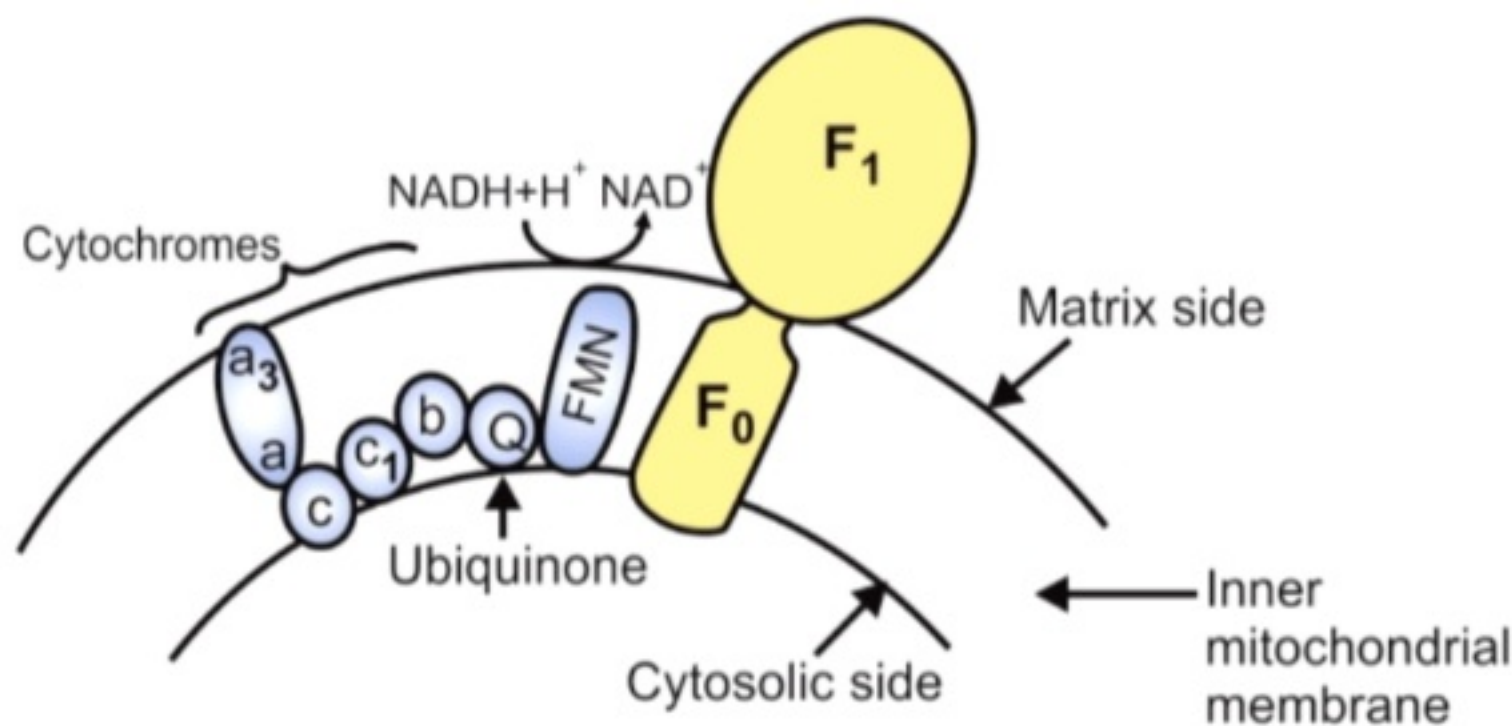


Figure 10.1: Structural organization of components of respiratory chain and F₀F₁ ATPase in the mitochondrial membrane

COMPONENTS OF THE ELECTRON TRANSPORT CHAIN:

The major components of the electron transport chain include:

- Nicotinamide adenine dinucleotide (NAD⁺).
- Flavin mononucleotide (FMN) and Flavin adenine dinucleotide (FAD).
- Ubiquinone or coenzyme Q.
- The **iron-sulfur (Fe-S) protein** associated with FMN and cytochrome b.

- **Cytochromes (hemeproteins):** b, c₁, c, a and a₃. Of these, only cytochrome c is water soluble and easily diffusible, whereas cytochromes b, c₁, a and a₃ are lipid soluble and therefore, are fixed components of the membrane. Cytochrome aa₃ are also called **cytochrome oxidase**; and are **copper containing hemeproteins**.

Except coenzyme Q, all members of this chain are proteins. Coenzyme Q (CoQ) is a fat soluble quinone (ubiquinone) and is a constituent of mitochondrial lipids.

STRUCTURAL ORGANIZATION OF COMPONENTS OF ELECTRON TRANSPORT CHAIN:

The mitochondrial electron carriers are organized into four complexes (complex I to IV) that catalyze oxidation-reduction reactions of the electron transport chain (Figure 10.2).

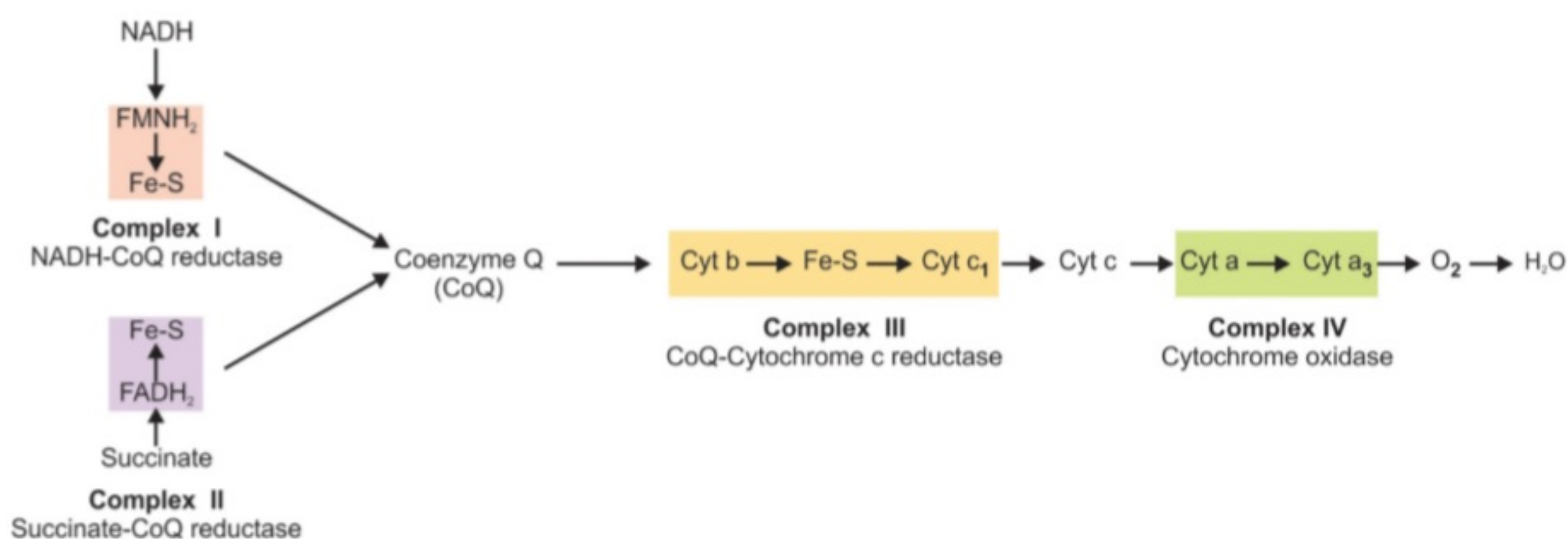


Figure 10.2: The electron transport complexes

- **Complex I, NADH- CoQ reductase**, catalyzes the transfer of electrons from NADH to coenzyme Q (CoQ).
- **Complex II, Succinate-CoQ reductase**, transfers electrons from succinate to coenzyme Q.
- **Complex III, CoQ- Cytochrome c reductase**, transfers electrons from CoQ to cytochrome c.
- **Complex IV, Cytochrome oxidase**, transfers electrons from cytochrome c to O₂.

Components of the respiratory chain are arranged in order of increasing redox potential (Table 10.1). Reducing equivalents flow through the chain from the

components of **more negative redox potential** to the components of **more positive redox potential**.

Table 10.1: The redox potential E_o' of redox couples of components of electron transport chain

<i>Redox couple components of electron transport chain</i>			<i>Redox potential E_o' in volt</i>
$2H^+ + 2e^-$	→	H_2	- 0.41
$NAD + H^+ + 2e^-$	→	NADH	- 0.32
$FMN + 2H^+ + 2e^-$	→	FMNH ₂	- 0.32
Ubiquinone + $2H^+ + 2e^-$	→	Ubiquinol	+ 0.04
Cytochrome b (ox) + e^-	→	Cytochrome b (red)	+ 0.07
Cytochrome c_1 (ox) + e^-	→	Cytochrome c_1 (red)	+ 0.23
Cytochrome c (ox) + e^-	→	Cytochrome c (red)	+ 0.25
Cytochrome a (ox) + e^-	→	Cytochrome a (red)	+ 0.29
Cytochrome a_3 (ox) + e^-	→	Cytochrome a_3 (red)	+ 0.55
$1/2O_2 + 2H^+ + 2e^-$	→	H_2O	+ 0.82

REACTIONS OF ELECTRON TRANSPORT CHAIN:

The following sequence of reactions occurs in the transfer of electrons from substrate to the ultimate acceptor oxygen (Figure 10.3).

1. NAD^+ is reduced to NADH by various dehydrogenases which remove two hydrogen atoms from their metabolite (MH_2) and get oxidized to M. In this oxidation reduction reaction, one hydrogen atom is accepted by NAD^+ to form NADH, while the second proton (H^+) is released into the aqueous medium.
2. The reduced NADH is oxidized by an enzyme **NADH dehydrogenase**. This enzyme contains **coenzyme FMN**. The coenzyme FMN accepts two electrons ($2e^-$) and a proton (H^+) from NADH and a free H^+ from the aqueous medium to form FMNH₂.
3. In addition to FMN, NADH dehydrogenase also consists of **Fe-S proteins**, which accepts only electron from FMNH₂. Thus two Fe-S protein molecules accept two electrons from one FMNH₂ molecule with release of two protons ($2H^+$) into the medium and FMNH₂ gets oxidized to FMN.

4. CoQ accepts two electrons from two Fe-S protein molecules and two protons ($2H^+$) from the medium to get reduced to $CoQH_2$. CoQ also collects reducing equivalents from $FADH_2$ formed by FAD-linked dehydrogenases.
5. Beyond CoQ, oxidation reduction process occurs by removal of electrons with the help of **cytochromes**. Cytochromes accept only electrons from coenzyme QH_2 with the release of $2H^+$ in the medium. As a cytochrome can accept only one electron, $CoQH_2$ transfers its two electrons to two molecules of cytochrome b, c1 c, a and a3 sequentially.
6. The last cytochrome complex is cytochrome oxidase (cyt aa3) which passes electrons from cytochrome c to molecular oxygen. Each oxygen atom accepts two electrons from cytochrome a3 and two protons from the medium and a molecule of water results.

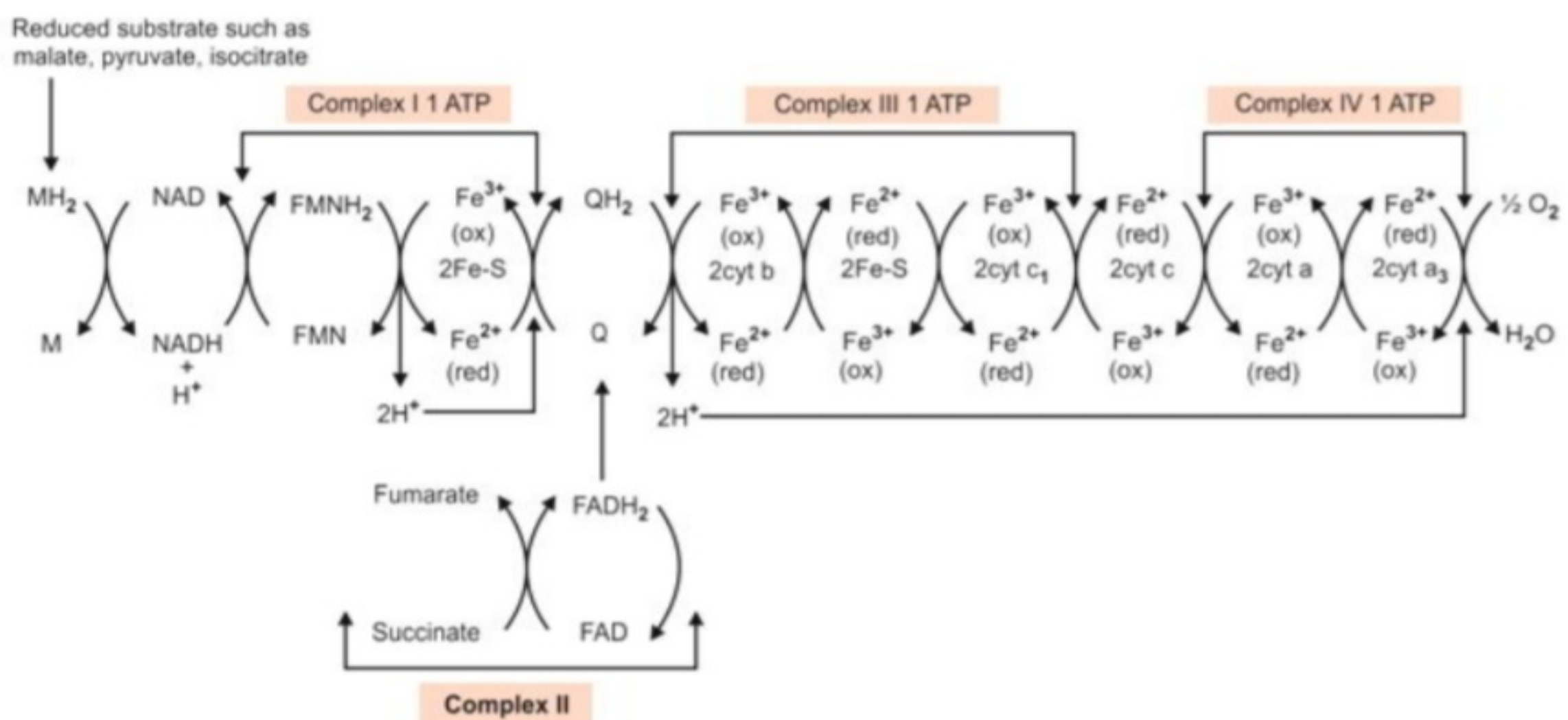


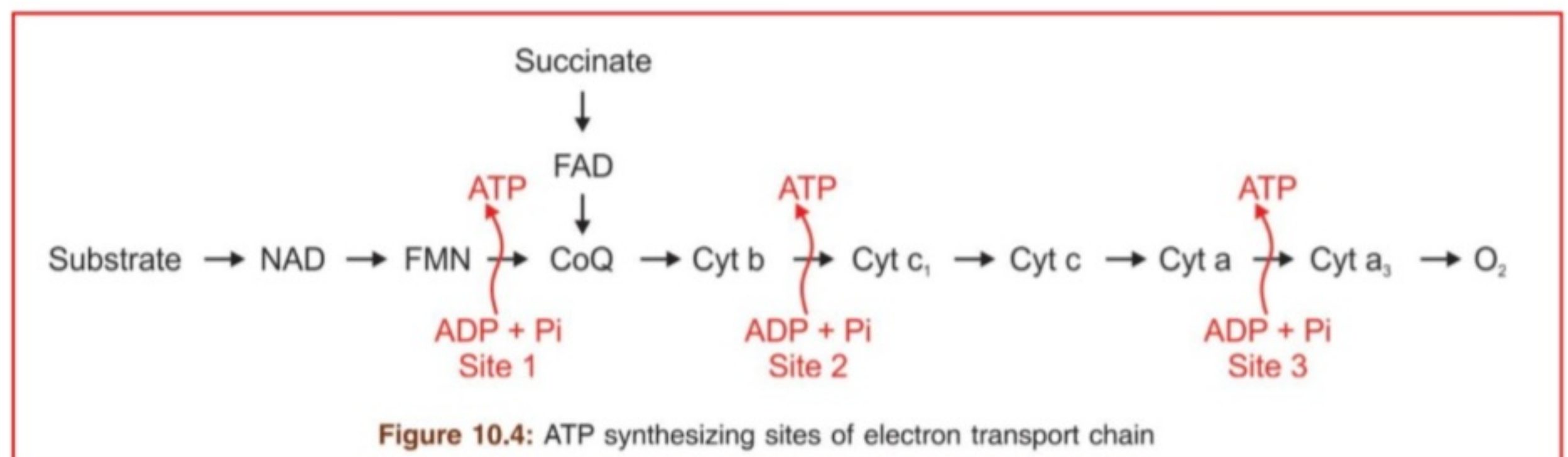
Figure 10.3: Electron transport chain

FORMATION OF ATP:

During the transfer of electrons through the electron transport chain, energy is produced. This energy is coupled to the formation of ATP molecules by phosphorylation of ADP by an enzyme **FoF₁ATPase**. The formation of ATP from ADP and P_i is termed **phosphorylation**, as phosphorylation is coupled with biological oxidation, the process is called **biological oxidative phosphorylation**.

SITES OF ATP SYNTHESIS:

- There are three ATP synthesizing sites of the electron transport chain, these are (**Figure 10.4**):
 - Oxidation of FMNH₂ by CoQ
 - Oxidation of cytochrome b by cytochrome c₁
 - Cytochrome oxidase reaction (oxidation of cytochrome a by cytochrome a₃).
- These sites provide the energy required to make ATP from ADP and Pi by an enzyme **F₀F₁ATPase**.
- Electrons that enter the chain through NADH pass through all three ATP synthesizing sites and thus yield three ATPs.
- However, electrons that enter the chain through FADH₂ pass through only two ATP synthesizing sites, as they bypass site 1, they yield two ATPs.



INHIBITORS OF ELECTRON TRANSPORT CHAIN:

Inhibitors of respiratory chain may be divided into three groups.

- Inhibitors of the electron transport chain proper
- Inhibitors of oxidative phosphorylation (**F₀F₁ATPase**)
- Uncouplers of oxidative phosphorylation.

1) INHIBITORS OF ELECTRON TRANSPORT CHAIN PROPER:

Inhibitors of electron transport chain proper include, inhibitors that inhibit the flow of electrons through the respiratory chain. These inhibitors block the respiratory chain at three sites:

1. Complex I (NADH to CoQ):

Inhibited by:

- Barbiturates such as amobarbital

- An antibiotic piericidin A
- The insecticide rotenone.

These inhibitors prevent the oxidation of substrates by blocking the transfer of reducing equivalents from Fe-S protein to CoQ.

2. Complex III (cytochrome b to cytochrome c1):

Inhibited by:

- Dimercaprol
- Antimycin A (antibiotics)
- British antilewisite (BAL), an antidote used against war gas.

These inhibitors prevent the transfer of electrons from cytochrome b to cytochrome C1.

3. Complex IV (cytochrome oxidase):

Inhibited by:

- Cyanide
- Carbon monoxide
- H₂S.

These inhibitors prevent transfer of electrons from cyt aa₃ to molecular oxygen by inhibiting cytochrome oxidase and can therefore totally arrest respiration.

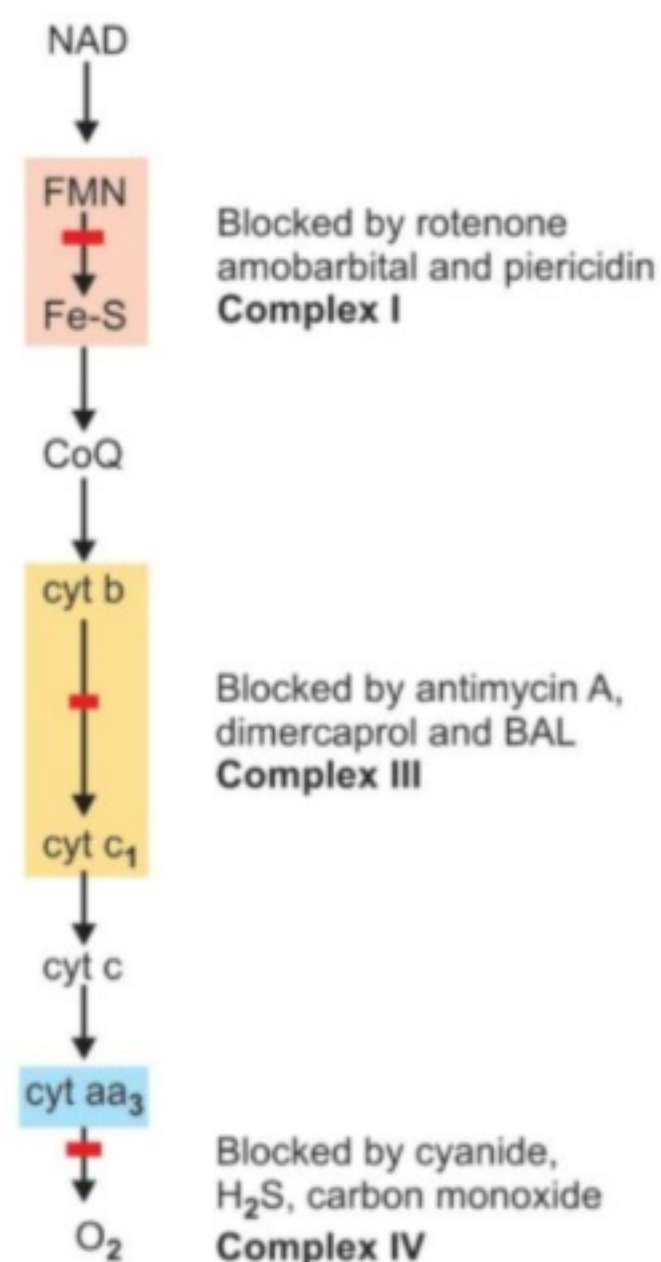


Figure 10.5: Sites of action of various inhibitors of electron transport chain

2) INHIBITORS OF OXIDATIVE PHOSPHORYLATION (F_0F_1 ATPase):

Another set of inhibitors do not inhibit special complexes. Instead, these compounds block phosphorylation directly by inhibiting **F_0F_1 ATPase** enzyme. For example, antibiotic **oligomycin** completely blocks oxidation and phosphorylation by inhibiting an enzyme F_0F_1 ATPase required for phosphorylation.

3) UNCOUPLERS OF OXIDATIVE PHOSPHORYLATION:

- Uncouplers are chemical substances that allow electron transport chain in mitochondria but prevent the phosphorylation of ADP to ATP by uncoupling the essential linkage between electron transport and phosphorylation for the synthesis of ATP.
- Uncoupling agents are lipophilic (lipid soluble) compounds, which readily diffuse through the mitochondrial membrane and are capable of binding H^+ ion.
- Uncouplers allow transport of H^+ ion across the membrane towards the side with the lower H^+ ion concentration, thus preventing the formation of proton gradient which is required for the formation of ATP.
- Thus, these compounds make the inner mitochondrial membrane abnormally permeable to protons. The energy produced by the transport of electrons is released as heat rather than being used for synthesis of ATP.

Examples of uncouplers include:

- 2,4-Dinitrophenol (DNP)
- Dicoumarol (an anticoagulant).
- Salicylate, a metabolite of aspirin.

PHYSIOLOGICAL UNCOUPLERS:

Certain physiological substances act as uncouplers, ex: thermogenin, thyroxine, bilirubin and free fatty acids. However, these compounds normally are not present in mitochondria in concentrations high enough to act as uncouplers.

IONOPHORES:

Ionophores means ion carrier molecules. They are lipid soluble substances, capable of binding and carrying specific cations (other than H^+) through the mitochondrial membrane. Oxidative phosphorylation can be prevented by certain ionophores. They differ from uncoupling agents in that they promote the transport

of cations other than H^+ through the membrane and abolish the membrane potential and/or pH gradient across the membrane and phosphorylation is therefore completely inhibited. For example, antibiotic **valinomycin** and **gramicidin**.



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